

FRED HOWARD PARK, FL, USA

WMO: 994008

Lat: 28.1533N

Lon: 82.8012W

Elev: 20

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 42.4	(c) 45.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 31.3	(k) 57.0	(l) 27.7	(m) 56.0	(n) 12.2	(o) 60	(p) 0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.4	(c) 89.5	(d) N/A	(e) 88.6	(f) N/A	(g) 87.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.5	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.2	(b) 22.1	(c) 19.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.8	59.6	62.9	65.9	72.0	77.3	81.8	82.4	83.1	81.0	75.6	67.3	63.8		
	DBStd	9.60	7.54	6.82	5.93	4.04	3.11	2.34	2.15	2.02	2.20	4.34	6.15	7.80		
	HDD50	23	14	2	1	0	0	0	0	0	0	0	0	6		
	HDD65	543	195	110	65	5	0	0	0	0	0	2	47	119		
	CDD50	8331	312	363	492	661	845	953	1004	1027	930	792	519	433		
	CDD65	3377	29	51	92	216	380	503	539	562	480	329	115	81		
	CDH74	30966	38	65	139	907	3075	5795	6309	6892	4961	2300	345	140		
	CDH80	9557	2	4	3	44	508	2005	2330	2759	1537	333	28	4		
(14) Wind	WSAvg	9.9	11.1	10.0	11.3	10.7	10.0	8.8	8.3	9.0	8.9	10.0	10.7	10.5		
(15) Precipitation	PrecAvg	53.0	3.0	3.3	3.9	1.9	2.8	6.1	7.9	8.4	7.2	2.9	2.1	3.1		
	PrecMax	80.2	8.0	10.9	14.7	4.9	15.2	18.3	22.2	18.5	18.1	14.1	9.1	15.6		
	PrecMin	36.7	0.1	0.3	0.0	0.0	0.0	0.6	0.9	2.6	0.4	0.0	0.0	0.0		
	PrecStd	9.5	1.7	2.4	3.1	1.5	3.0	3.6	3.9	3.2	3.9	2.8	1.9	3.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.5	79.1	80.0	83.3	87.3	90.4	90.8	91.2	90.3	87.1	83.0	80.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	74.6	75.6	77.2	81.2	85.5	88.9	89.0	89.4	88.8	84.7	80.1	77.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	72.4	73.9	75.7	79.9	84.1	87.5	87.7	88.5	87.2	83.1	77.8	75.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	70.3	71.9	74.3	78.5	82.8	86.5	86.8	87.5	86.1	82.0	75.7	73.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	10.2	10.7	9.7	8.9	8.3	8.2	8.4	9.8	10.3	11.4	11.1		
		MCDBR	13.0	10.7	11.9	11.6	11.5	9.7	9.3	9.3	11.6	11.1	12.4	11.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.357	0.368	0.392	0.425	0.451	0.476	0.459	0.423	0.390	0.354	0.343		
	taud		2.523	2.495	2.454	2.354	2.319	2.322	2.261	2.323	2.391	2.466	2.511	2.553		
	Ebn at Noon		285	290	293	287	276	267	260	265	272	275	279	281		
	Edh at Noon		29	32	35	40	42	41	43	41	37	32	29	27		
(44) All-Sky Solar Radiation	RadAvg		1109	1335	1707	2056	2191	1986	1872	1797	1633	1504	1218	1023		
	RadStd		80	95	152	109	120	157	111	112	117	95	71	86		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		+0.77	N/A	N/A	N/A	N/A	N/A	-7	-88	+301	+209

Nomenclature: See separate page